

EN 621: Design of Energy Systems

Course Code	EN 621
Course Name	Design of Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	MTech Energy (Mechanical)
Distribution	Elective for MS
Semester	Odd/Even
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Preamble

Starting from the background of foundation courses regarding energy systems this core course will focus on detailed design of conventional as well as emerging (hydro, thermal power plants including solar as well as fossil fuel powered, renewable power plants i.e. wind power plants etc.) plant system components and their integration. Functionality of components, design requirement and challenges will be covered in detail.

Course Outline

The objective of the course is to introduce the postgraduate students to advanced topics in Energy system designs. The course aims to provide the students an understanding of the fundamental concepts and techniques, used in the design, performance analysis, and implementation of current Energy systems and useful in the development of the renewable and non-renewable energy conversion devices of the future.

Course Modules

- Introduction to energy systems, Introduction to critical components for design in energy system, thermal stresses, material selections (4)
- Design of pressure vessels, Stresses in pressure vessels, Autofrettage, Thermal stresses, Design of pressure vessel components such as shell, heads, nozzles, flanges as per ASME & IS codes, Buckling (6)
- Design of high pressure power plant piping systems, flow characteristics, material selection, thermal analysis of pipe and joints, thermal insulations (4)
- Design of material handling in power plants, coal conveyor belts, ash handling system, slurry disposal system, handling of radioactive materials (8)
- Design of heat transfer equipments, Boilers: classification, selection, Heat exchangers: classification, selection, flow friction and pressure drop analysis, basic thermal design, e-NTU, p-NTU, MTD methods. Shell and tube heat exchanger, construction and thermal features, thermal design procedure, Kern method, Bell Delaware method (8)
- Thermal design of regenerators, classifications, design parameters. Design of compact heat exchangers, plate and fin, fin-tube and plate and frame heat exchangers, fouling and corrosion in heat exchangers (6)
- Wind turbine blade design, overview of design criteria and certification guidelines, aerodynamic design, structural design, design and choice of sub-systems and components (6)

References

1. J. F. Harvey, *Theory and Design of Pressure Vessels*, CBS Publishers and Distributors, 1987.
2. S. Walas, *Chemical Process Equipment, Selection and Design*, Butterworths Sr. in Chemical Engineering.
3. L. Brownell, E. Young, *Process Equipment Design*, John Wiley and Sons.
4. D. Kern, *Process Heat Transfer*, Tata McGraw-Hill, 2000.
5. Fraas, *Heat Exchanger Design*, Second Edition, John Wiley & Sons, 1989.
6. J. Manwell, J. McGowan, A. Rogers, *Wind Energy Explained*, Theory, Design and Application, Wiley, 2012.
7. R. Kulwiec, *Materials Handling Handbook*, ASME, John Wiley and Sons.
8. W. Stoecker, *Design of thermal systems*, Tata McGraw-Hill Education, 3rd Edition.